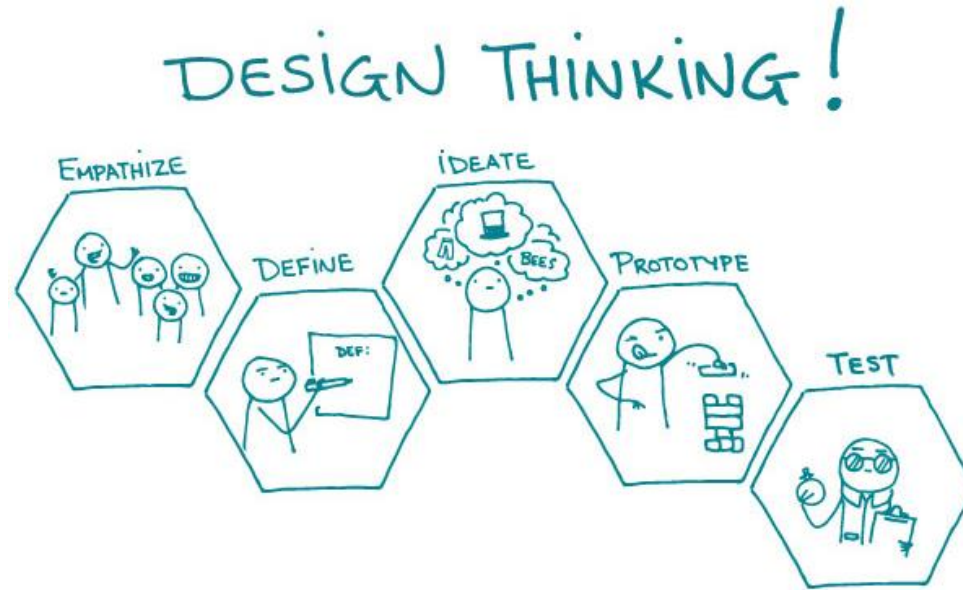




Design Thinking: The Complete Process

Term 4, EPD

March 2026

What is Design to YOU?



Think of how you
would define
DESIGN in a
simple way!



Design vs. Design Thinking: The Difference

DESIGN  design thinking



individuals



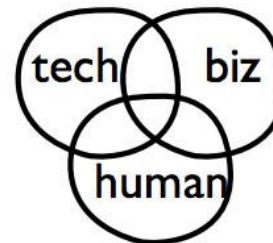
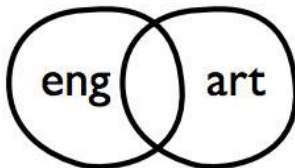
teams



products



experiences



Source: Stanford Design School

Design Thinking Leads to “Innovation”

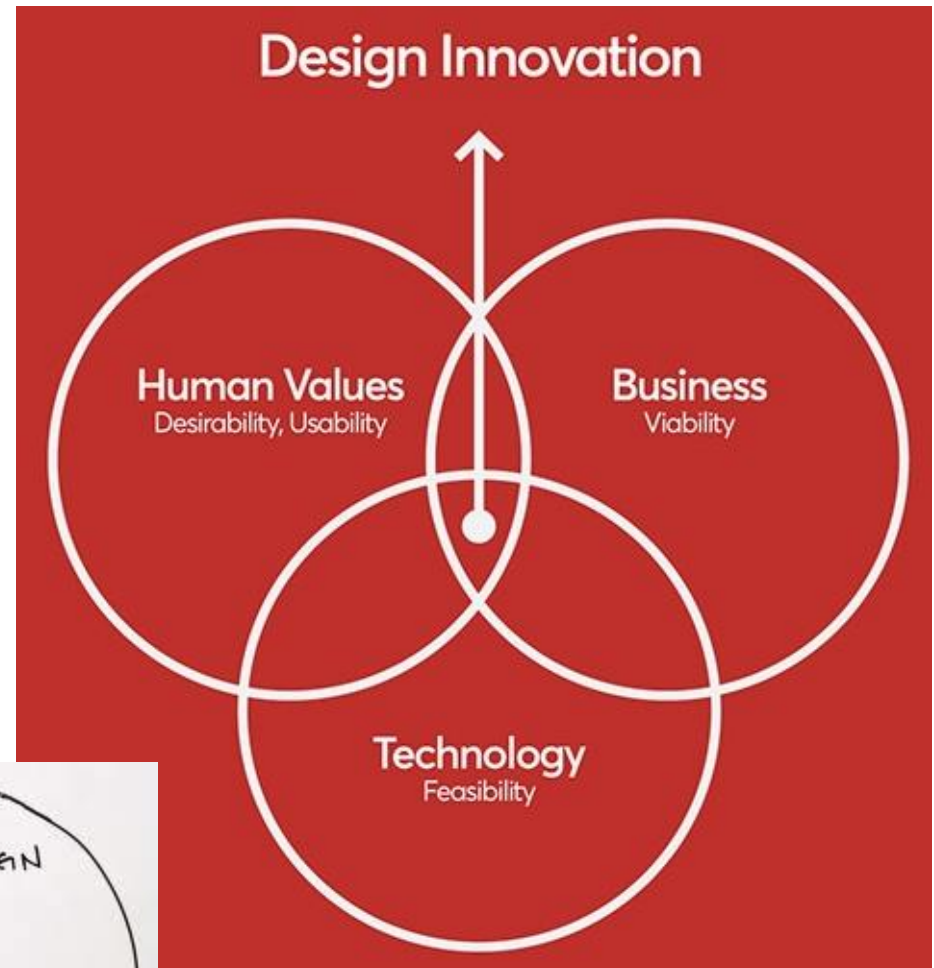
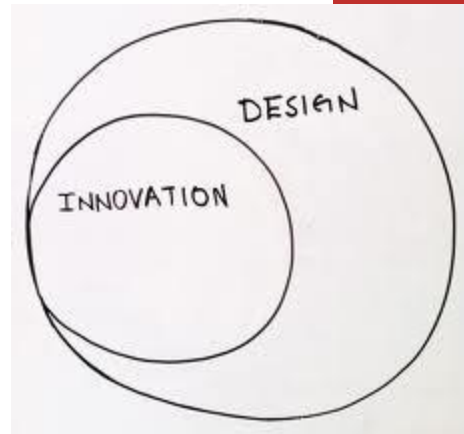
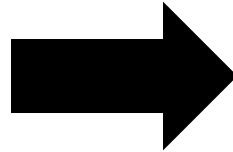
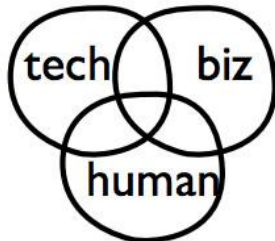
design **thinking**



teams



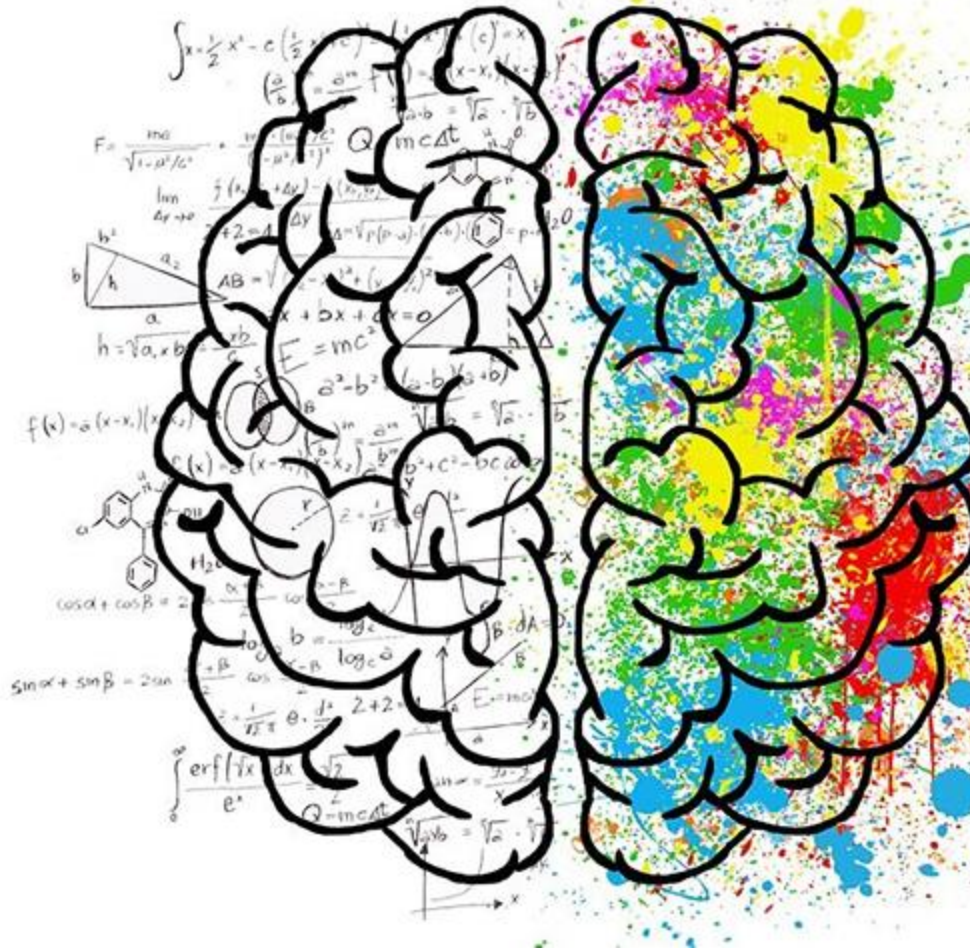
experiences



Design Thinking and Your Brain

Design Thinking stimulates your **right brain!**

LOGIC



CREATIVE

Before “Design Thinking” → Remember This:

**We're not
our users.**

An authentic focus on people begins with this simple acknowledgment.

IBM Design Research

Why Design Thinking?

IDEO: Why Design Thinking?



What?

Design Thinking is a methodology for creating or improving products, services and experiences grounded by the needs of the user rather than the whims of the creator.

Developed by IDEO, taught at the Stanford d.school.

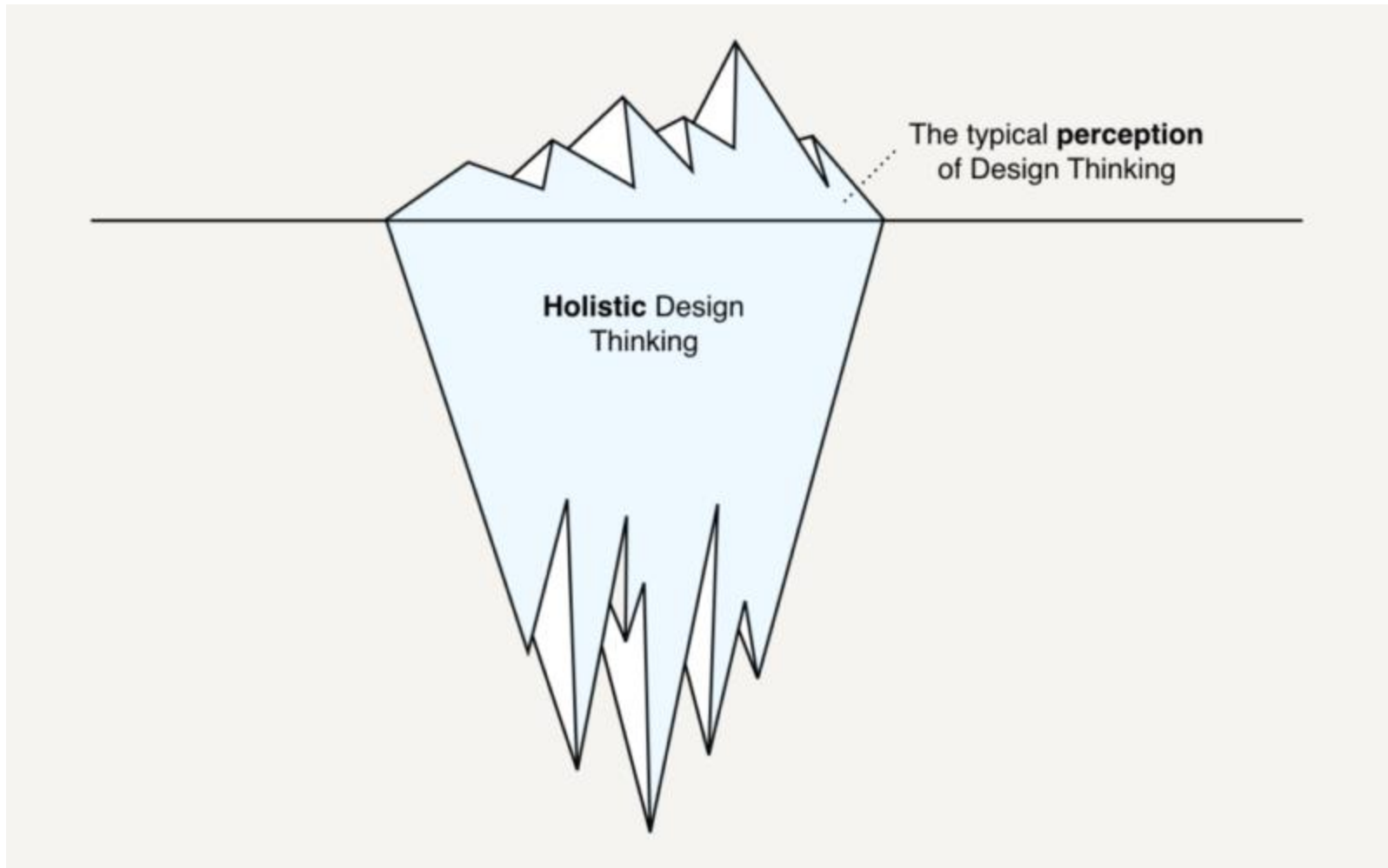
Who?

IDEO, GE, IBM, Capital One, Fidelity, Intuit, SAP, Liberty Mutual, Lego, Con Agra, Lowes Hardware, AirBnB, YOU!

Why?

Empathy drives innovation.

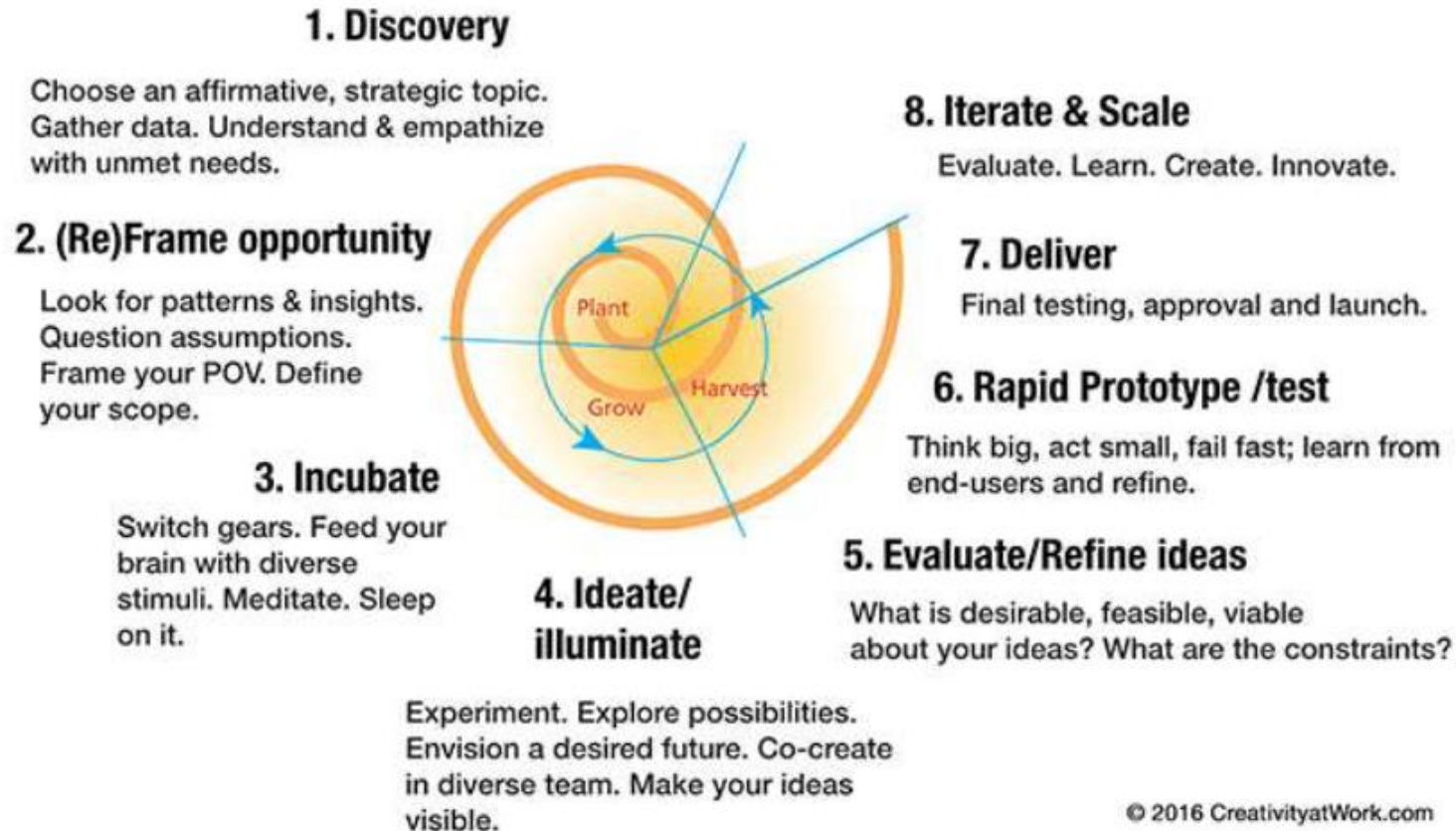
Perception of Design Thinking



Framework for Design Thinking

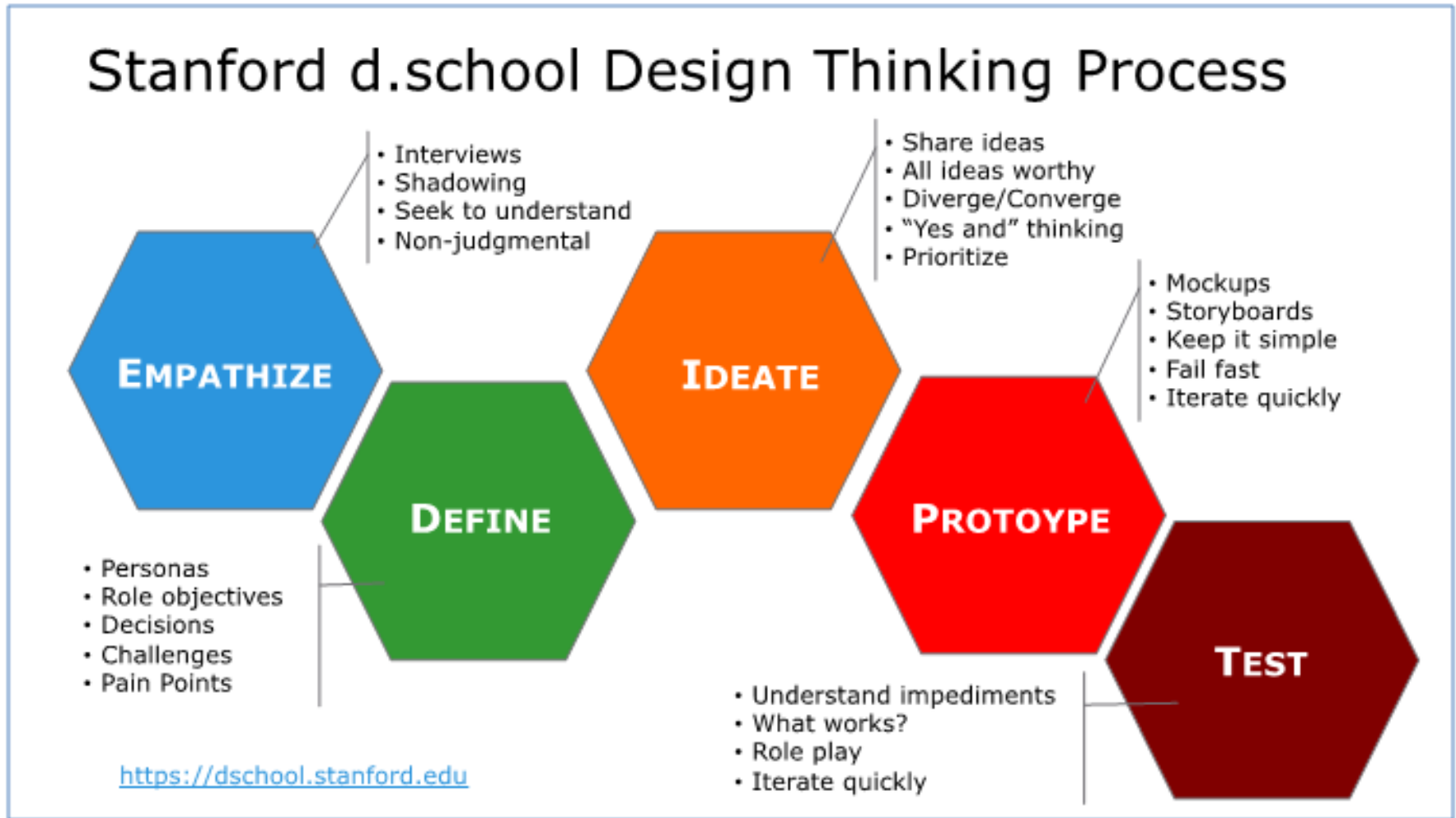
A Framework for Design Thinking

Continuous improvement and optimization of the solution is a critical aspect for the success of Design Thinking.



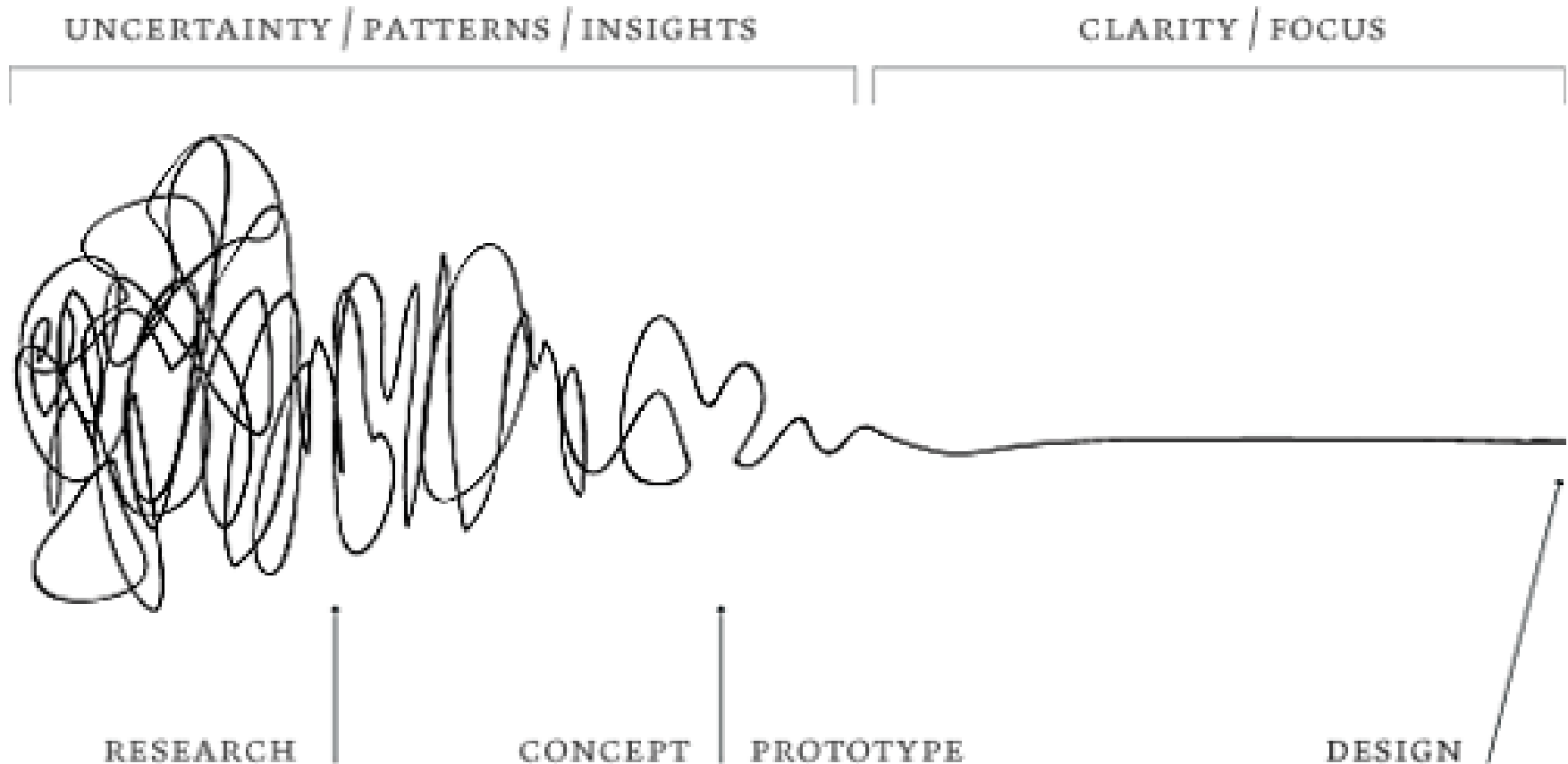
© 2016 CreativityatWork.com

Design Thinking – Stanford Design School Approach

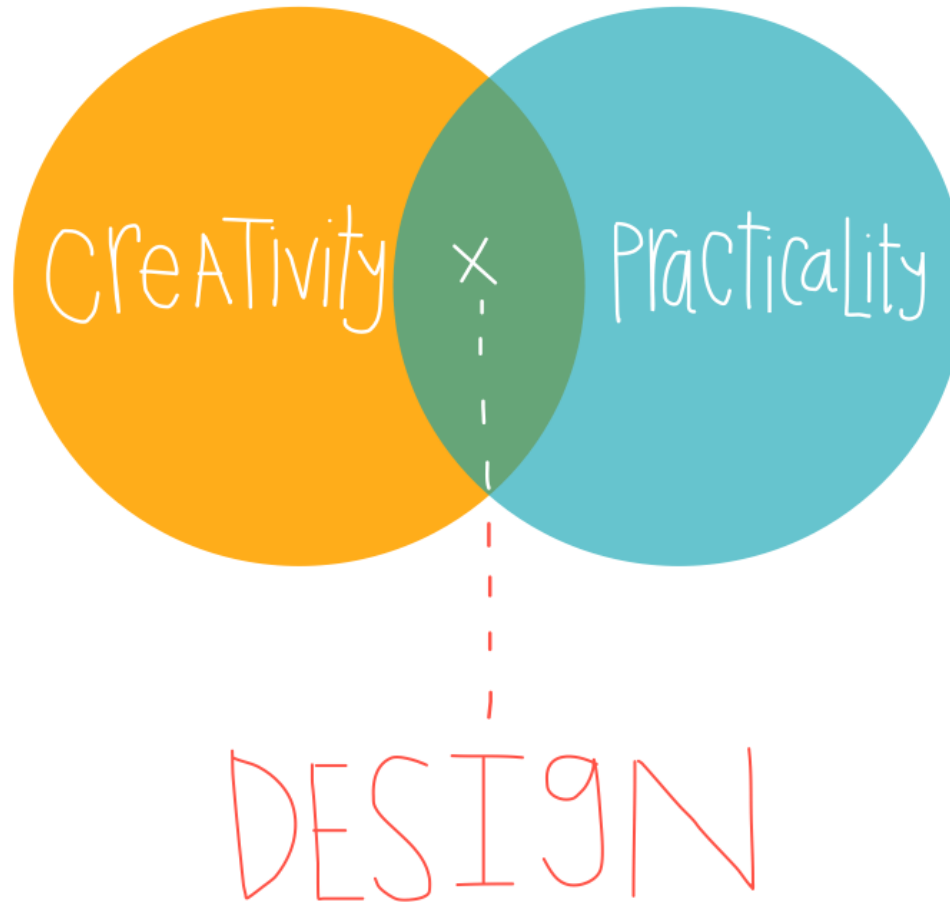


<https://dschool.stanford.edu/>

Design Thinking: From Chaos to Clarity



Final Goal of a Design Process



Stage 1: Empathize



- Do a lot of background research and review.
- **Information Gathering:** Mentors, users, manufacturers, government agencies, research articles, newspapers, white papers, pamphlets, site visits, surveys, organized workshops, interviews.
- Learn about the problem given to you.
- What is the broad problem that you are trying to solve?
- Do not make judgements at this stage. Do not filter out information and knowledge gained. Keep mind open to note down all views and opinions and facts and figures.

Stage 2: Define



STAGE 2

- Identify the **needs**, **wants** and **demands** of the various stakeholders in your project.
- List out all the **challenges** to be addressed and **constraints** within the project (technical, legal, financial, environmental etc...)
- Problem framing – Define the **problem statement** – what problem exactly are you going to solve? Why are you solving it? How is it going to improve the current situation or way of life? What evidence do you have to back up and support your problem? How are you going to approach the problem moving forward? What might your solution look like?

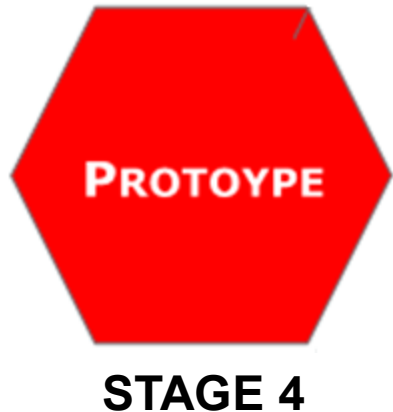
Stage 3: Ideate



STAGE 3

- **Brainstorming - Concept generation.** Come up with crazy ideas and sketches and “*think wild*” about all types of solutions that you want to come up with to solve the problem.
- Your “concepts” could be in the form of sketches, flow charts, small scale models etc... Do not ask yourself whether the concept is going to work or not.
- Find inspiration from the world around you in coming up with “crazy concepts”. No limit to how many concepts you come up with – the more, the better.
- **Concept Evaluation** – Rank the various concepts in terms of feasibility, cost, robustness etc... Identify 2 or 3 strong concepts worth taking further.

Stage 4: Prototype



- **Test out your ideas “quickly”**. Check whether they can work and they are realistic.
- Make good use of *simulation softwares, mathematical models, paper mock-ups, small scale prototypes using off-the-shelf cheap hardware / software, open source codes etc...*
- Do this for the 2 or 3 strong ideas you have identified. These ideas should be clearly “distinct” in their approach and idea.
- Once **basic “functionality check”** is done for these multiple ideas, identify the “strongest” one and go ahead full on to do **in-depth prototyping**.

Stage 5: Test

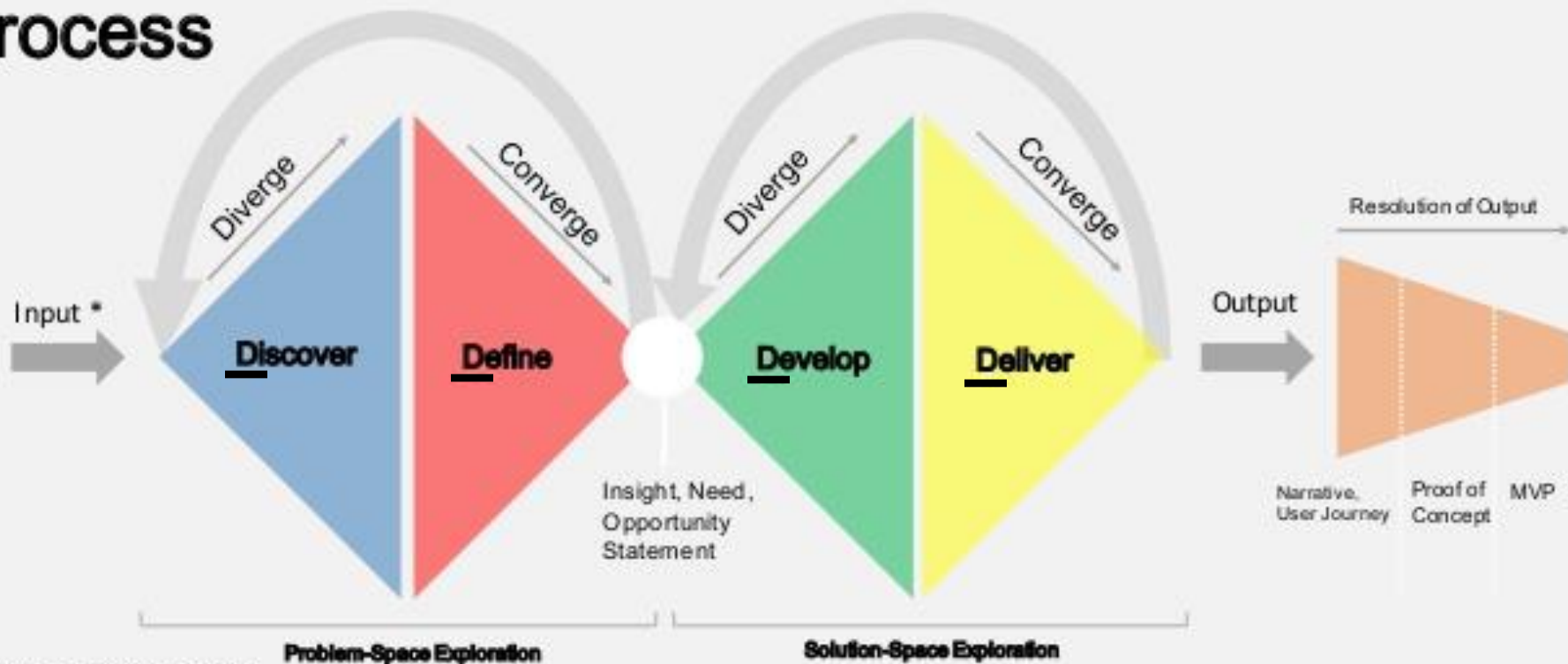


STAGE 5

- Build your full scale prototype (you run a low risk of it not working because of Stages 1-4).
- Test it out for **functionality** under different use case scenarios.
- Iterate within the prototype by modifying components or sub-systems to improve performance and functionality.
- Check for “**robustness**” of your solution (stress testing) and tweak system to make it more robust (choice of right materials, sensors etc...). Always fall back on all the background research you have done to guide your full scale prototyping efforts.
- Do be prepared for any “**risks**” so that you can quickly prototype your 2nd best solution, if need be.

Design Thinking: Diverge & Converge

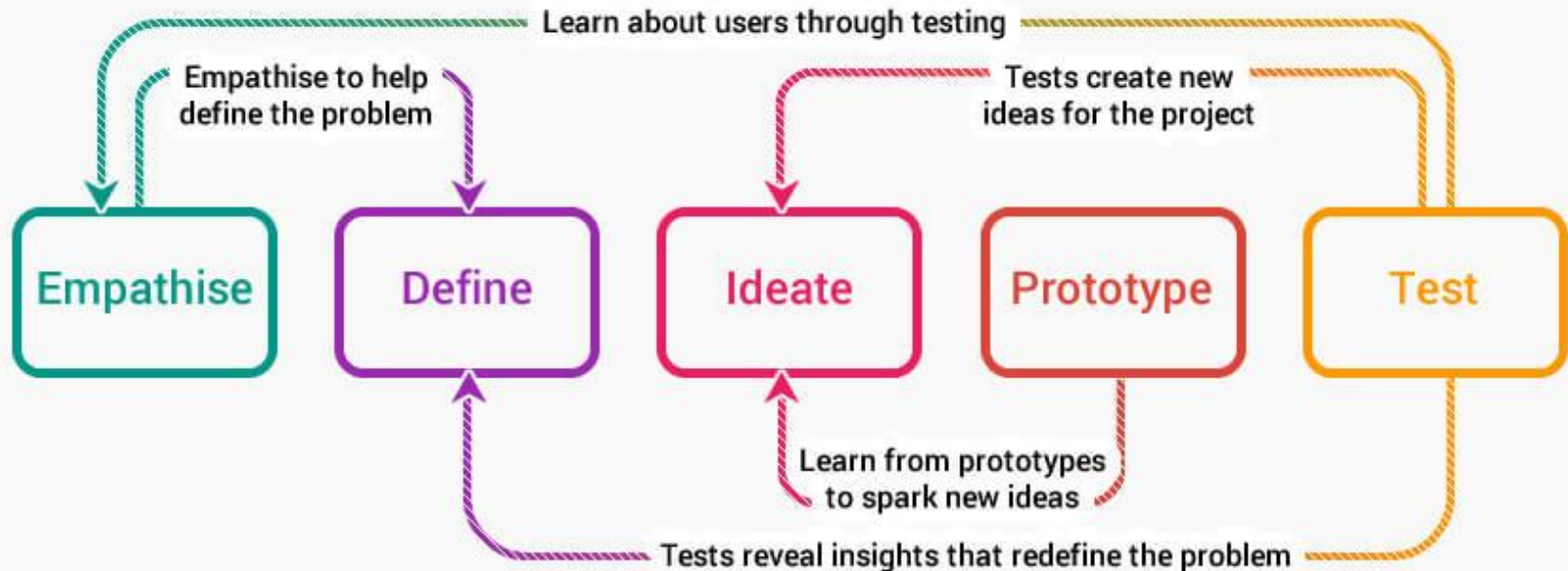
Design Thinking Process



*Ambiguous/Complex Problem
OR
Greenfields (purposefully
vague brief to find new and radical ideas)

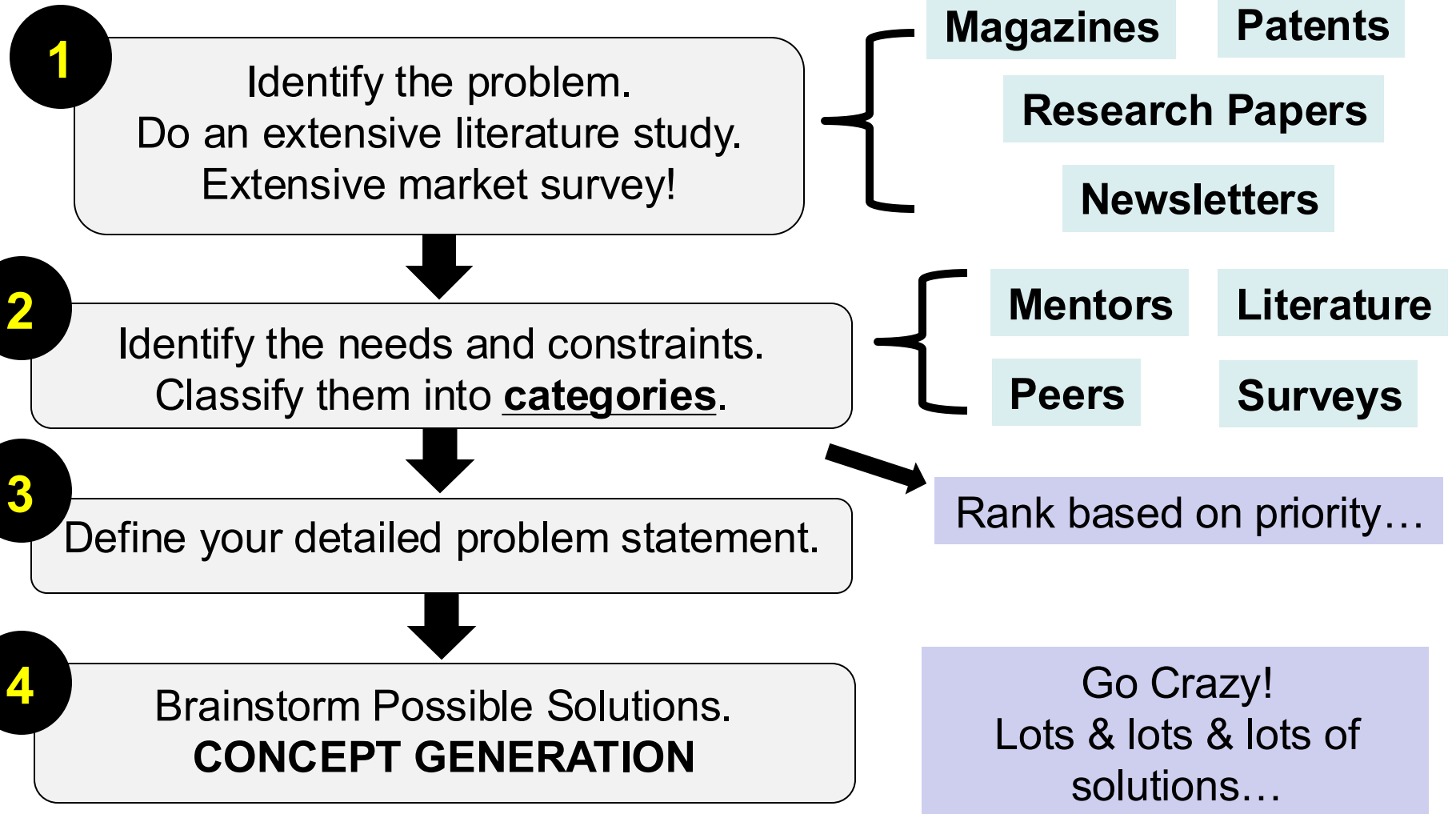
Design Thinking: It's a **Non-Linear** Process

DESIGN THINKING: A NON-LINEAR PROCESS



Source: interaction-design.org

The Overall Product Design Cycle (1/2)



The Overall Product Design Cycle (2/2)

5

CONCEPT EVALUATION

Rank the different design solutions.

Metrics to assess the quality of your solution.

Team Feedback

Surveys

Technical Basis

Non-Technical Basis

Mentor Feedback

6

PROOF-OF-CONCEPT (PoC)

Simple Prototyping of Best Solutions.
Modeling / Simulations.



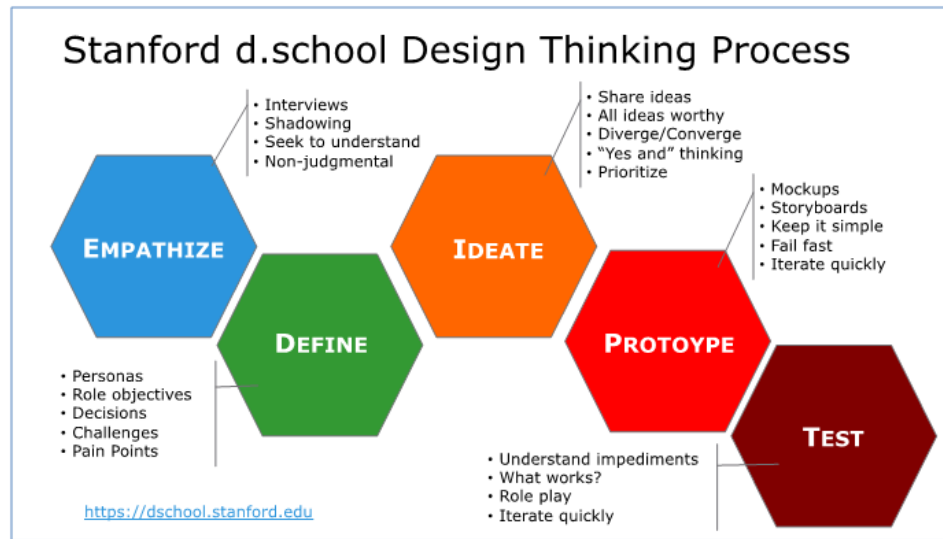
7

FULL FLEDGED PROTOTYPING

Building and Testing.
Evaluating the prototype.
Multiple iterations to improve functionality, usability and robustness.

$$\begin{cases} \frac{dI}{dt} = \frac{\alpha_I}{1 + \left(\frac{R}{R_{50}}\right)^{\mu_R}} + \frac{\alpha_A}{1 + \left(\frac{A}{A_{50}}\right)^{\mu_A}} \cdot e^{\beta \cdot UVc} - \delta \cdot I & (1.1) \\ \frac{dR}{dt} = \frac{\alpha_R}{1 + \left(\frac{I^F}{I_{50}^F}\right)^{\mu_I}} - \delta \cdot R & (1.2) \\ I^F = \frac{I}{1 + \left(\frac{IPTG}{IPTG_{50}}\right)^2} & (1.3) \end{cases}$$

So what if I don't follow this process?



You will still be able to solve the problem.

Yes, BUT...

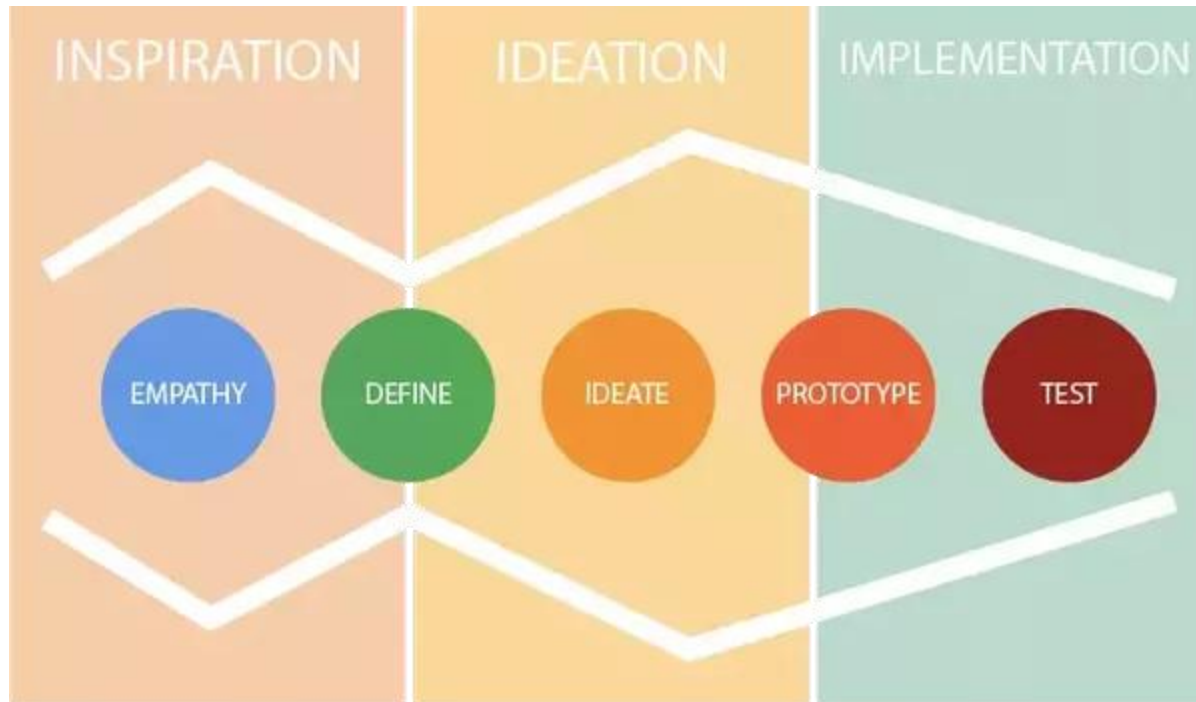
Was that really a problem? Is that why current products were not being sold enough?

Does your solution have a market value? Do customers really need what you did "great"?



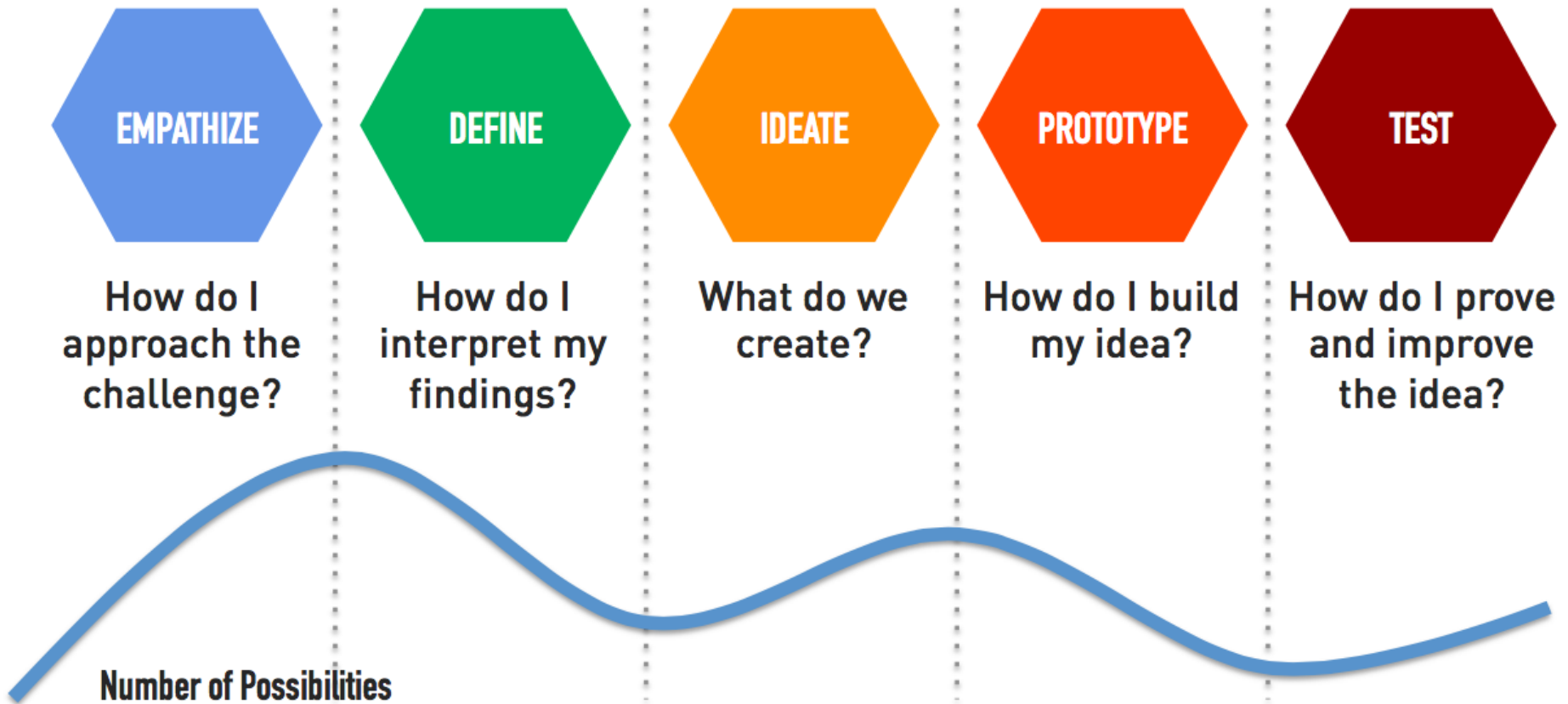
Maybe you will realize too late that some one has already done what you did after 7 months of your efforts.

Advantages of Using “Design Thinking”

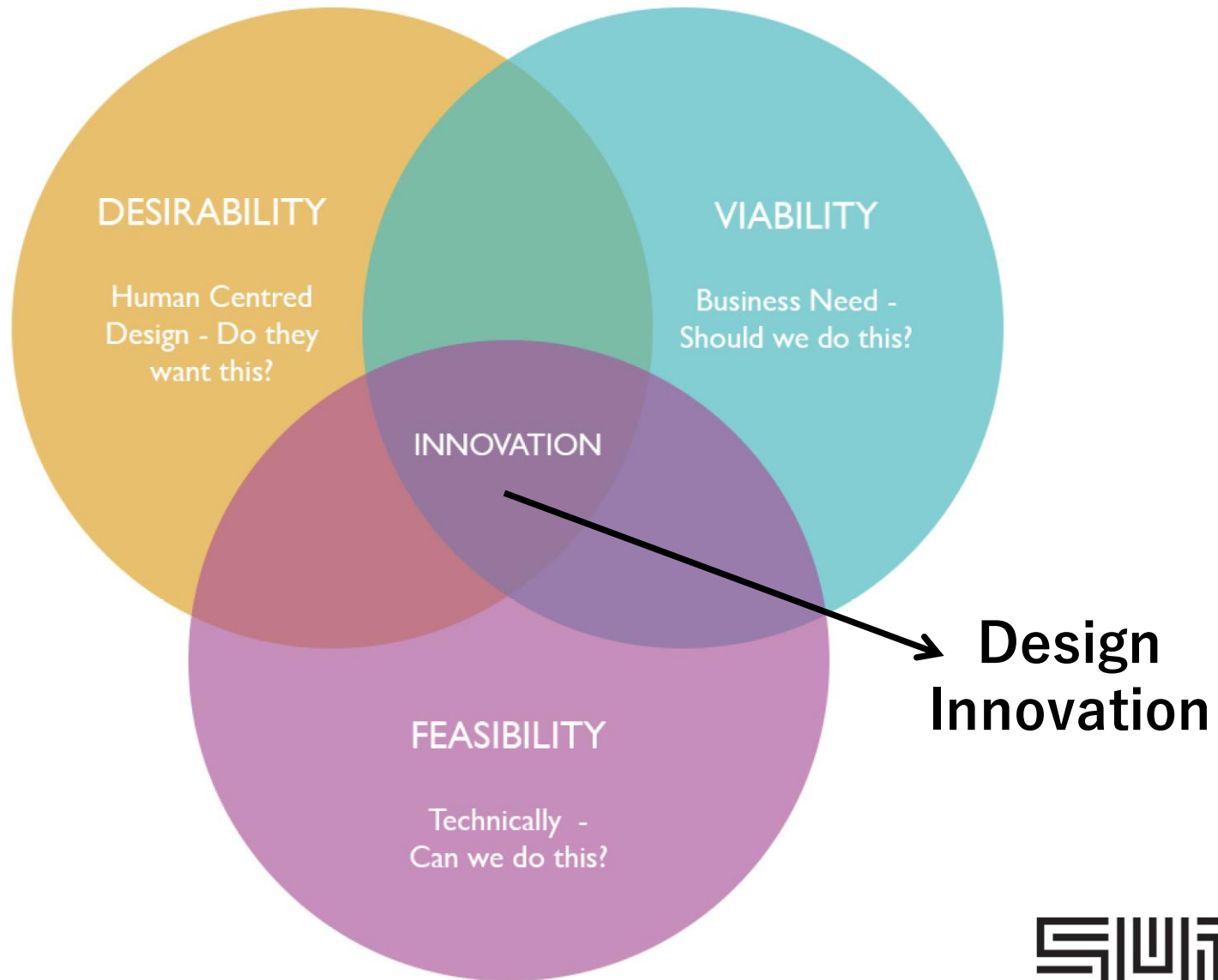


- Confidence boost. New insights to you and to the company.
- Low risk of product failure.
- High customer perceived value.
- Financial success and market creation / capture.
- More novel solutions. Blue ocean strategy.

Design Thinking Process: What Should I Ask Myself?



Design Thinking: Venn Diagram



Design Thinking Process: Different Figures, Same Thought!

DESIGN THINKING FRAMEWORK



IMPLEMENT

Put the vision
into effect



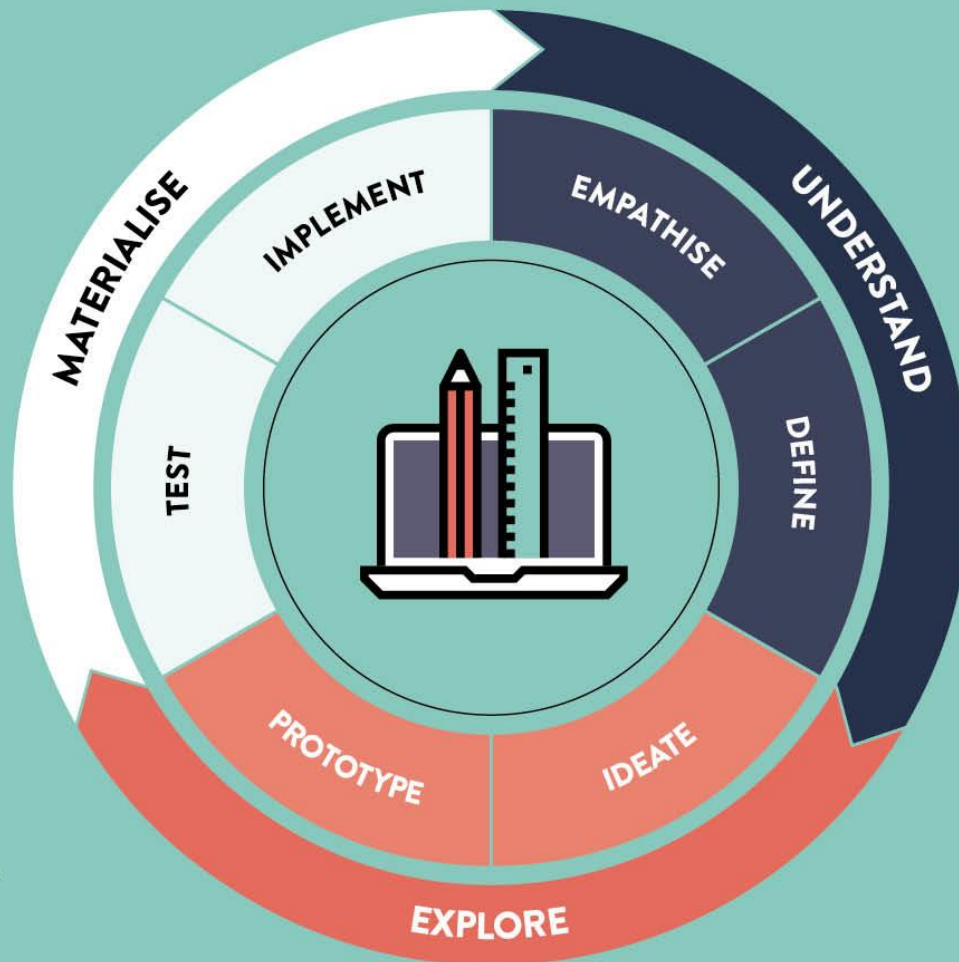
TEST

Return to
your users
for feedback



PROTOTYPE

Build real, tactile
representations
for a range of your ideas



EMPATHISE

Conduct research
to develop an
understanding of
your users



DEFINE

Combine all your
research and
observe where your users' problems exist



IDEATE

Generate a
range of crazy,
creative ideas

Source: Nielsen Norman Group

The AEIOU Framework

- AEIOU is a framework that **guides designers** in **thinking through a problem** or scenario from a **variety of perspectives**:
 - **A**ctivities
 - **E**nvironment
 - **I**nteractions
 - **O**bjects
 - **U**sers
- They are useful in **organizing** thoughts, observations, and ideas into distinct categories.

A

Activities are goal-directed sets of actions. What are the pathways that people take toward the things they want to accomplish, including specific actions and processes? How long do they spend doing something? Who are they doing it with?

E

Environments include the entire arena where activities take place. For example, what describes the atmosphere and function of the context, including individual and shared spaces?

I

Interactions are between a person and someone or something else, and are the building blocks of activities. What is the nature of routine and special interactions between people, between people and objects in their environment, and across distances?

O

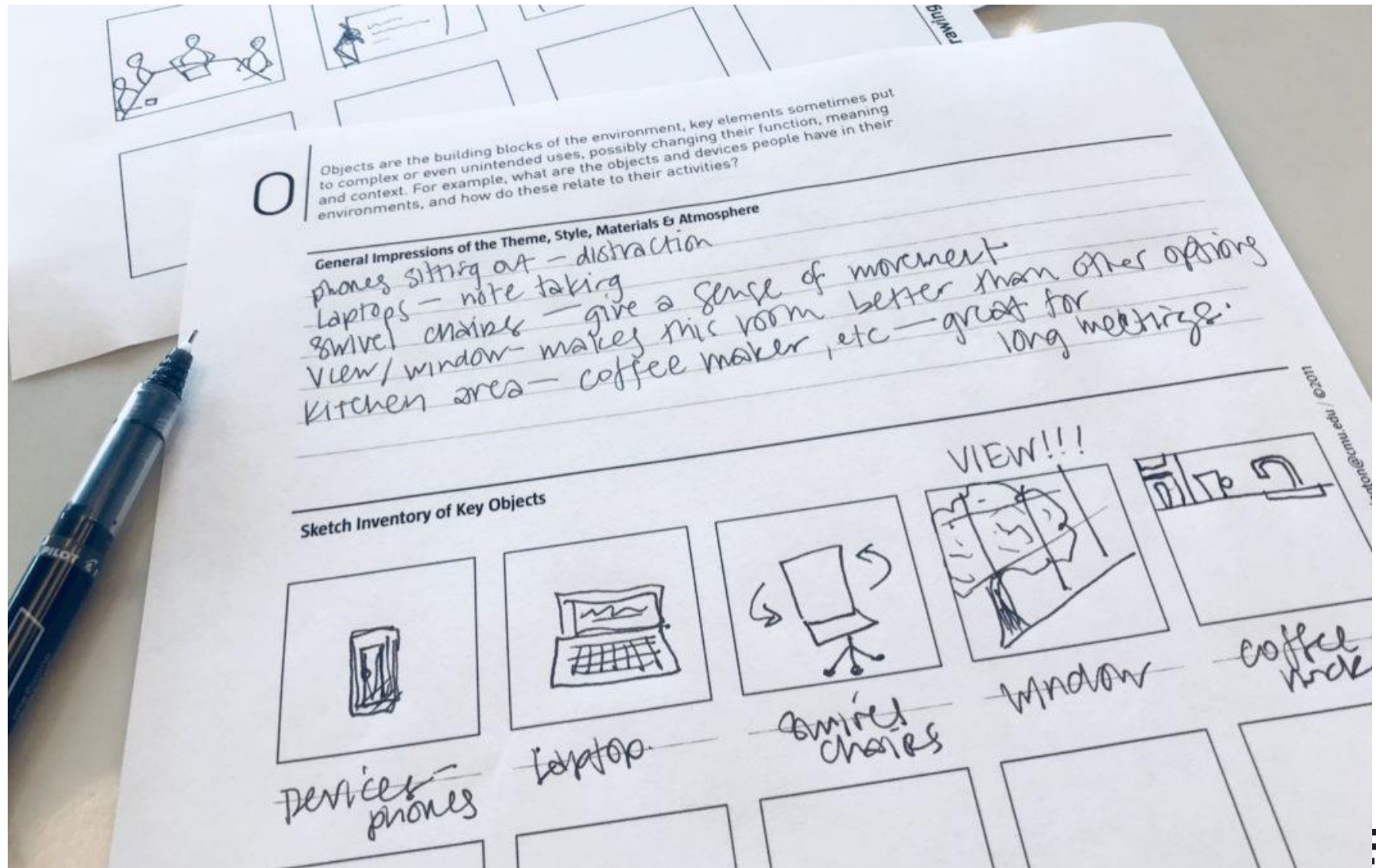
Objects are the building blocks of the environment, key elements sometimes put to complex or even unintended uses, possibly changing their function, meaning and context. For example, what are the objects and devices people have in their environments, and how do these relate to their activities?

The AEIOU Framework

U

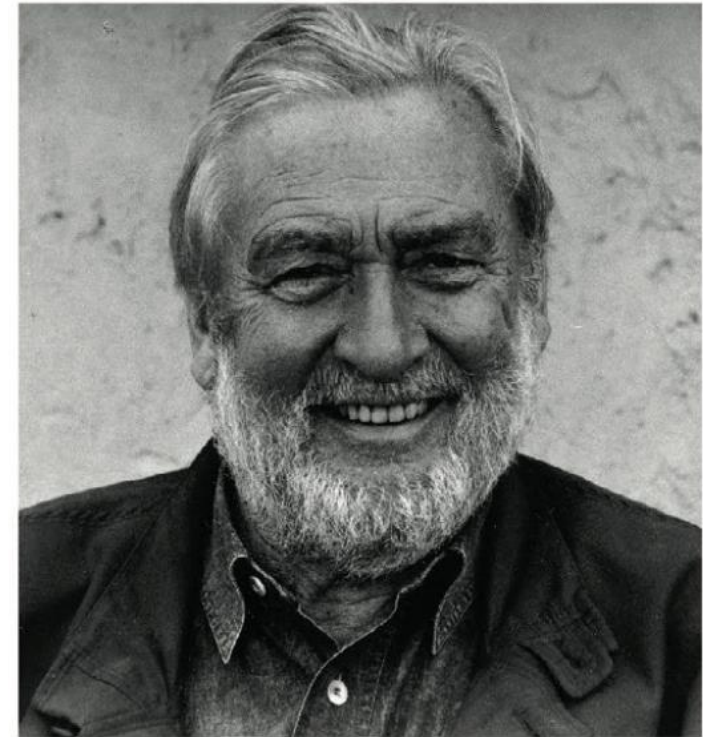
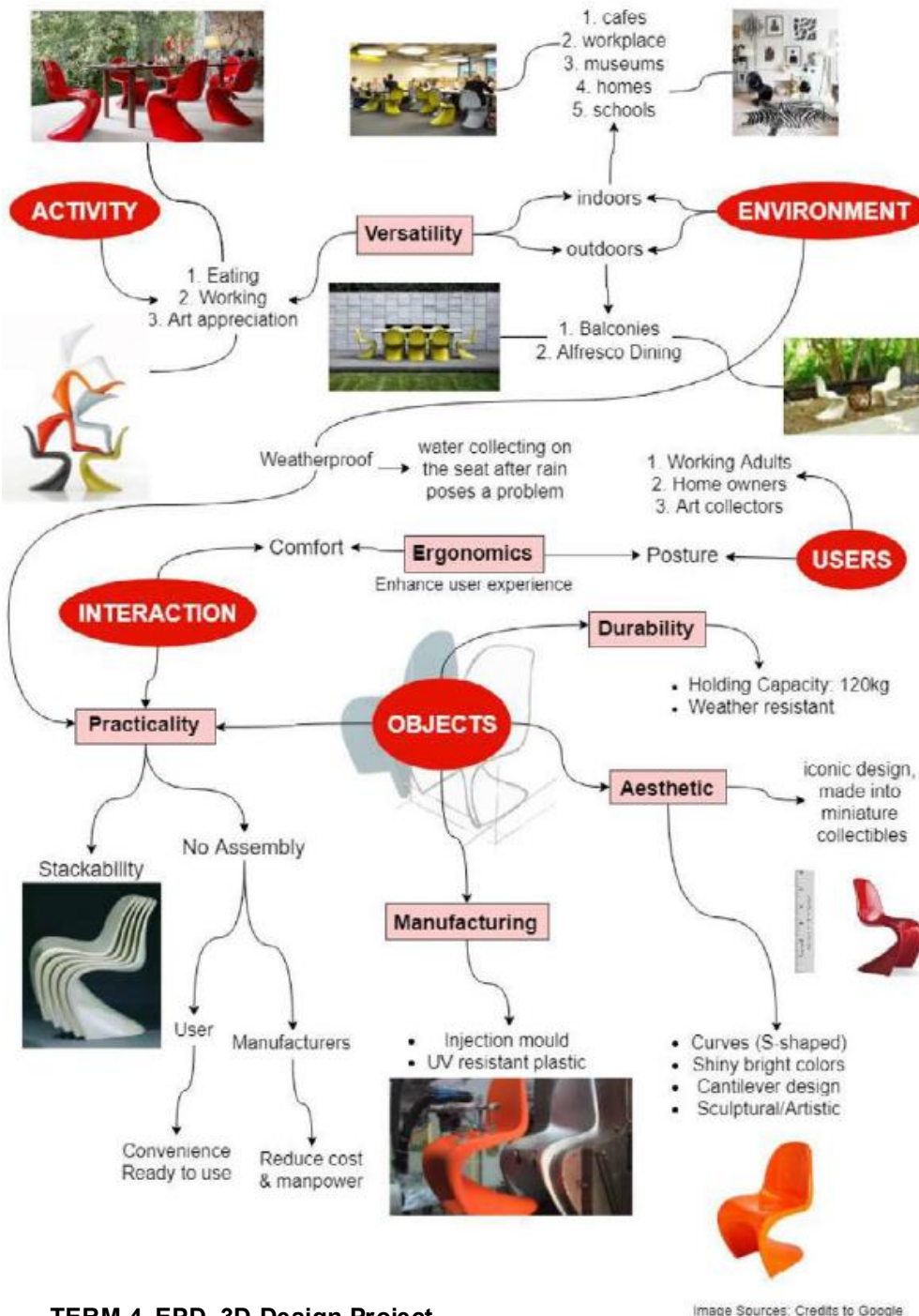
Users are the people whose behaviors, preferences, and needs are being observed. Who is present? What are their roles and relationships? What are their values and prejudices?

Use the AEIOU Worksheets (if you want...), Available on E-Dimension!





Examples of AEIOU Framework



Verner Panton (1926-1998)

Panton Chair Verner Panton, 1967

Post WWII, the injection-moulding industry boomed due to an increased need for inexpensive mass-production. This brought about the introduction of injection-moulding into the furniture industry. With the impetus of using such technology and breaking away from the norms of wood-based furniture, Verner Panton decided to create the Panton Chair, a chair that boasts a cantilevered, S-shaped and legless design, all while being produced from a single, continuous piece of Polypropylene. Verner Panton crafted an iconic piece of furniture that preceded his time, leaving behind a lasting legacy of original, intelligent design which pioneered the injection-moulding technology in the furniture industry.

Environment

Private spaces seem to be the preferred choice for people to cut their nails, as most consider this task as one that should be done in private. Below are some potential places where people choose to cut their nails.



General Requirements

- Good lighting for visibility
- Elevated surfaces to accommodate sitting position
- Collection or disposal points
- Storage points
- Privacy

Object Lists

- Lamp, Room Lights
- Chair, Table, Stool, Bed
- Waste Bin, Scrap / Tissue Paper
- Drawing Bag or Container

Inappropriate Environment

Shared spaces
eg. Public transportation, waiting benches

- Socially frowned upon
- Unhygienic
- Inconsiderate
- Limits the environment



<https://imgur.com/rf1y9w6>
<https://vredd.it/59y9f7433l.png>

Activity

Intended use

- Cutting nails
- Basic nail detailings
- Manicure, Pedicure

Unintended use

- Cutting wires
- Cutting threads

User

Personal Use

- Children
- Teenagers
- Adults

Service

- Caretakers
- Pet Groomiers
- Manicurist
- Pedicurist

Interaction

User Operation of Nail Clipper



User retrieves nail clipper from storage and twists it open

Cutting Process (Finger)



1. Set up collection point by laying down a piece of tissue paper
2. Operate nail clipper with one hand
3. Align clipper to obtain desired cut, maintaining force while cutting

Filing Process (Finger)



1. Twist out the nail file
2. File the nail for finer detailing

Cutting Process (Toe)



Before repeating the earlier cutting steps, reposition as each:

1. Set down
2. Elevate foot on stool
3. Lean forward to have better visibility while cutting

(When done it is to maintain safety and comfort while cutting the nail)

Cleaning Up / Disposal Process



User holds up tissue paper with clipped nail inside and disposes of them accordingly

Examples of AEIOU Framework



Nail Clipper

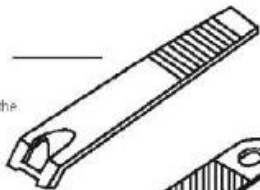
Patented in 1875, Valentine Fogerty



Objects

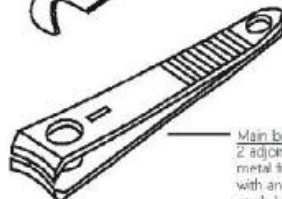
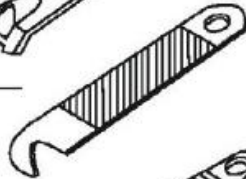
Lever

A point of contact for the user to apply force to compress the cutting component on the main body.



Nail digger

Contains a filing component and a sharp, curved tip. It effectively reaches to smaller and tighter areas of the nail to uncover dirt and enables the user to also polish and shape their nails.



Main body:
2 adjoining pieces of metal fixed together with an inclined angle between them. It is the main cutting element of the nail clipper.



Joining pin

This component brings the lever and the main body together and compresses the adjoining metal pieces on the main body for the cutting process.

Some Inspiring Design Quotes!

I HAVE
NOT FAILED.
I'VE JUST
FOUND 10,000
WAYS THAT
WON'T WORK.

THOMAS A. EDISON

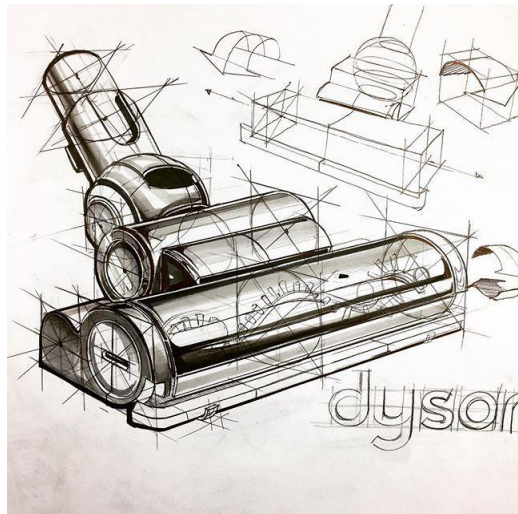
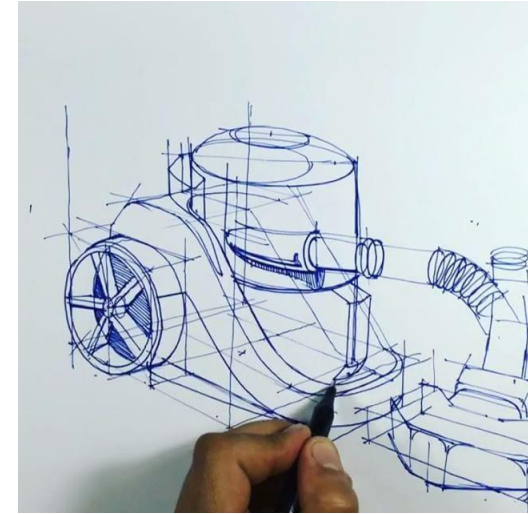
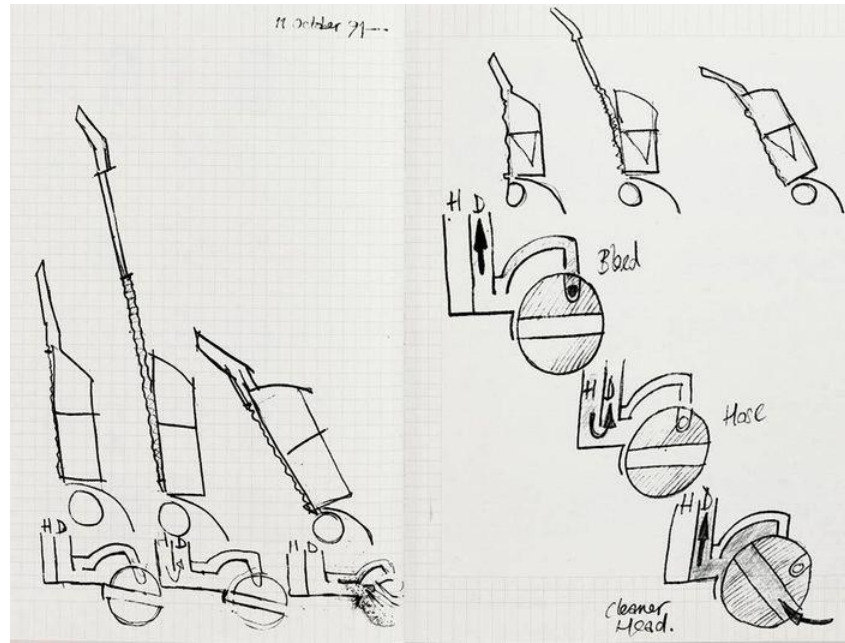
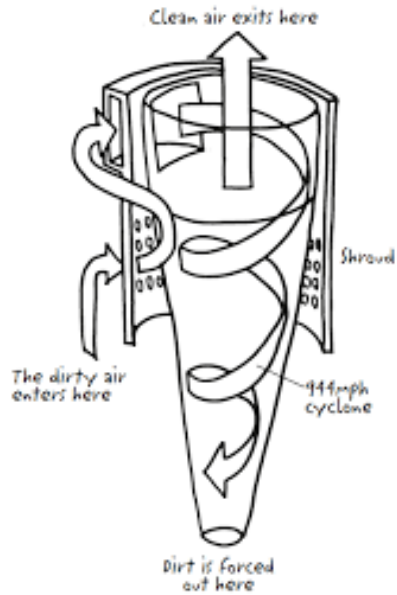
jennifer secher | StudioJR

**GOOD IDEAS CAN
COME AT ANY TIME,
SO ALWAYS HAVE A
PENCIL TO HAND.**

I made 5,127 prototypes of
my vacuum before I got it
right. There were 5,126
failures. But I learned from
each one. That's how I
came up with a solution. So
I don't mind failure.

James Dyson

DYSON's Design Sketches...

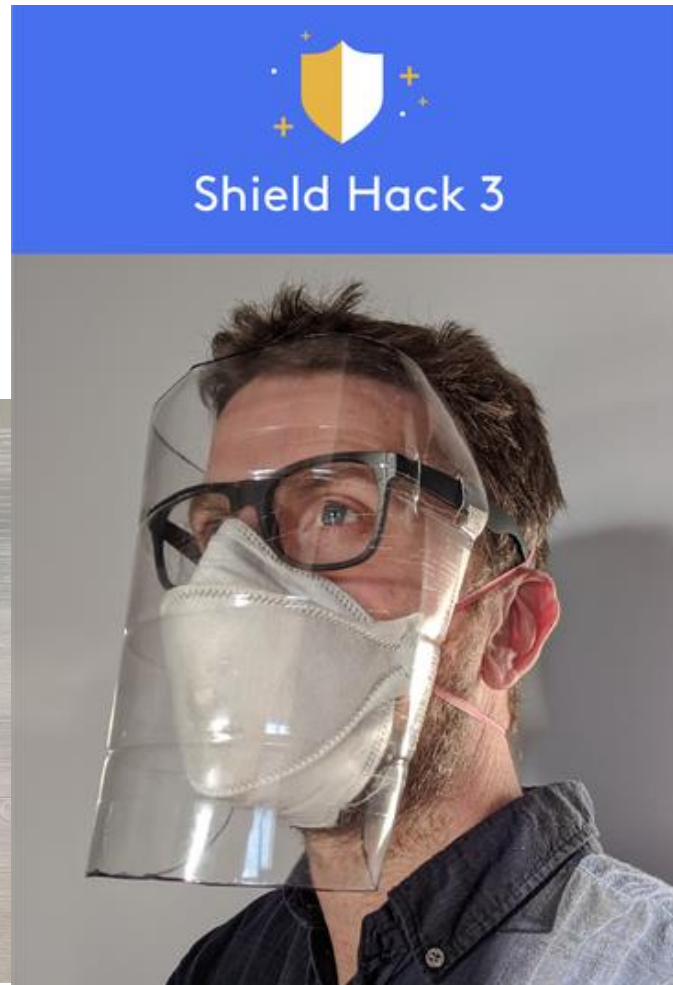


dyson



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

Collaborative Design Thinking in the COVID Era



INSTRUCTIONS:

1



4



2



5



3



6



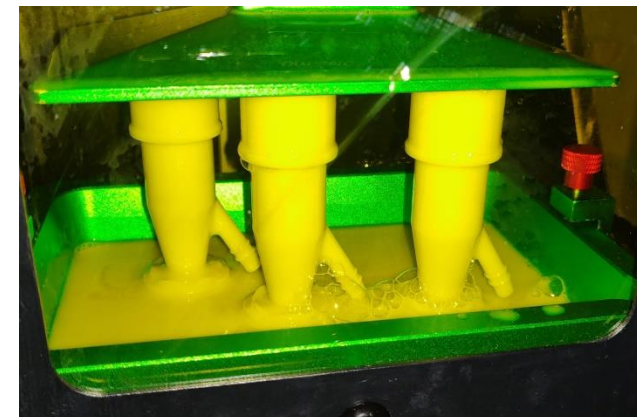


Photo taken in Lombardy, ITALY shows 3D-printed respirator valve parts used to transform a common snorkeling mask into an emergency respiratory mask

Short Lectures to Come...

- Topic 1 – Design Thinking: The Complete Process
- Topic 2 – Identifying Customer Needs and Constraints
- Topic 3 – Problem Framing
- Topic 4 – Concept Generation
- Topic 5 – Concept Selection



Useful Readings



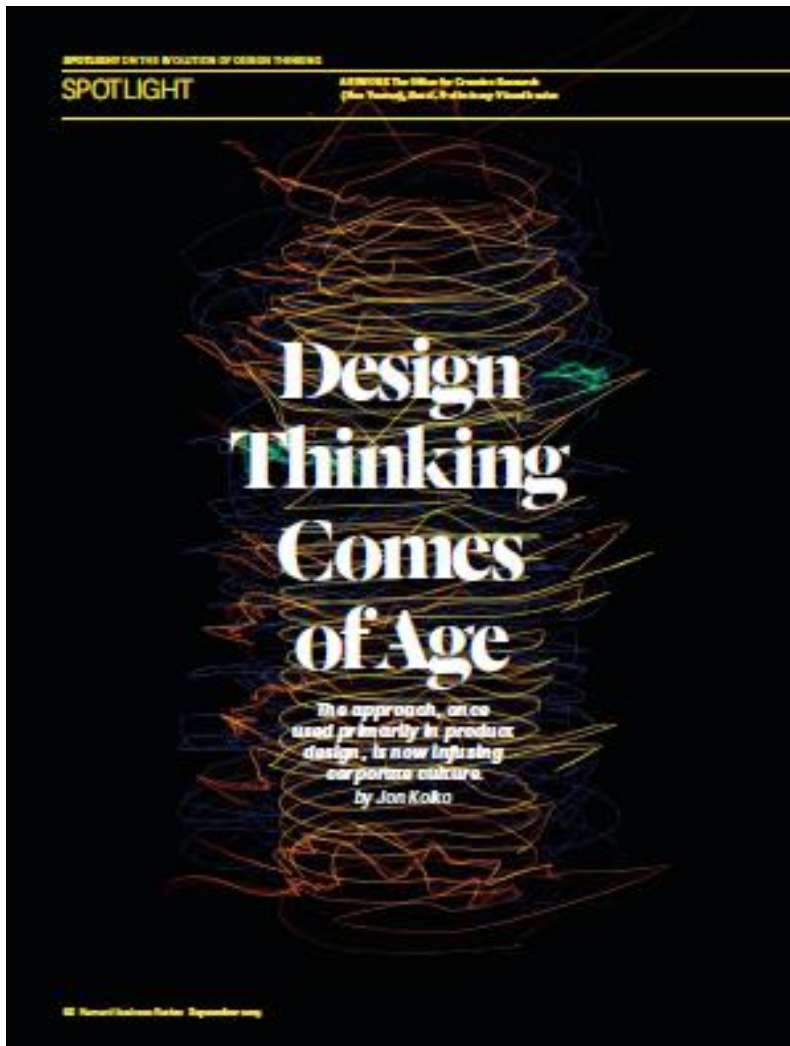
Harvard Business Review
Sept-Oct 2018, Page 72

Most criticism of design thinking comes from a misunderstanding of its purpose. Design thinking is not just a tool for designers; it's one that nondesigners can use to come up with creative solutions to problems. Designers criticize it because it undermines their expertise: If anyone can solve design problems, why do we even need designers?

As a designer, I can see how design thinking can hurt the design industry. But I don't think design should be limited to a select few. It would be wonderful to see everyone practice it, and design thinking provides a solid framework to do so.

Michael Degtyarev,
senior UX/UI designer, Yardi

**Harvard
Business
Review
Press**



There's a shift under way
in large organizations,
one that puts design
much closer to the
center of the enterprise.

**Design thinking is an essential tool for
simplifying and humanizing. It can't be
extra; it needs to be a core competence.**



**Harvard
Business
Review
Press**

**Harvard Business Review
June 2008, Page 85**

Useful Readings

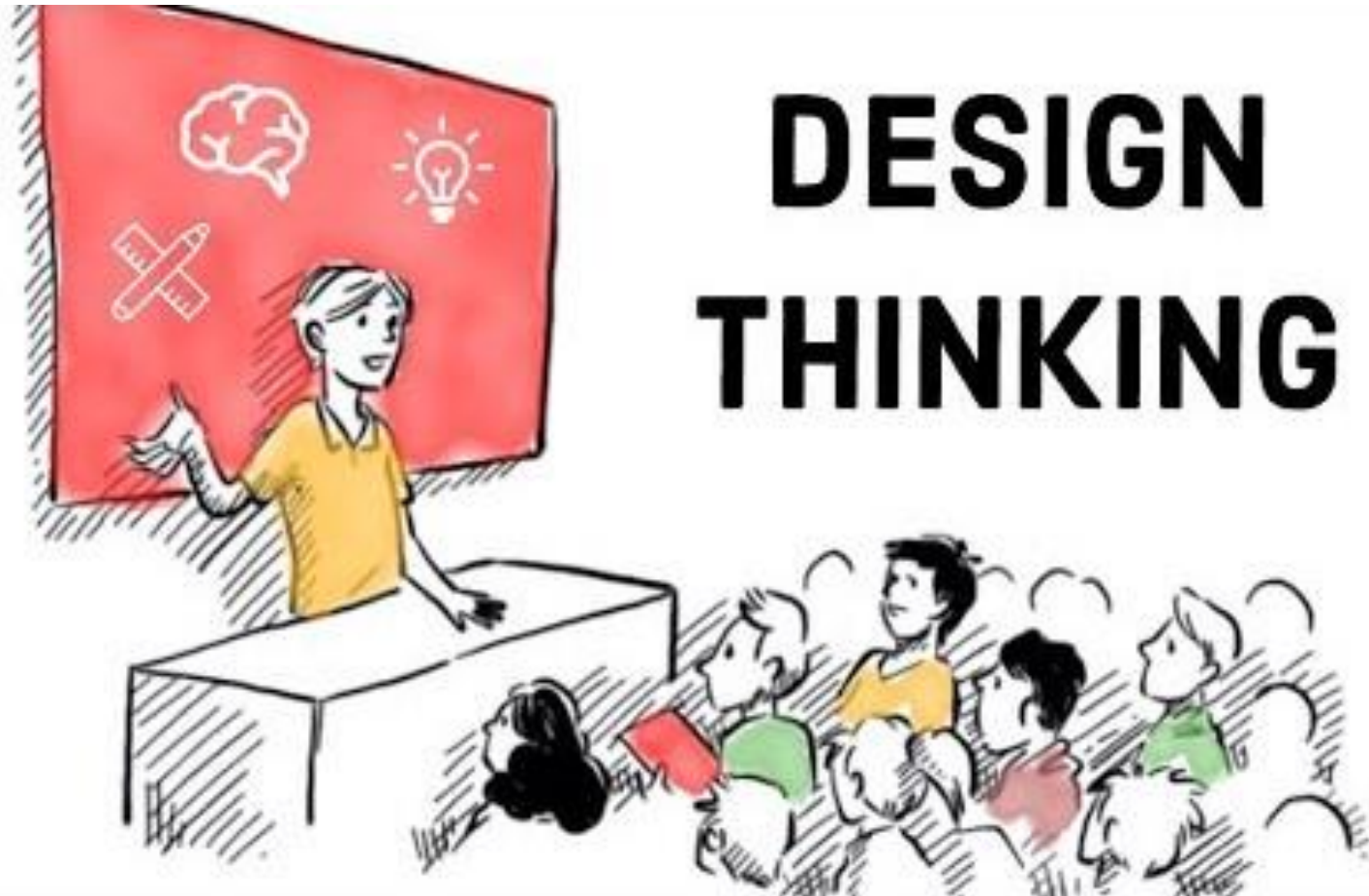
**A Design Thinker's
Personality Profile**

Empathy. They can imagine the world from multiple perspectives – those of colleagues, clients, end users, and customers (current and prospective). By taking a “people first” approach, design thinkers can imagine solutions that are inherently desirable and meet explicit or latent needs. Great design thinkers observe the world in minute detail. They notice things that others do not and use their insights to inspire innovation.

The surgeons described a new device for sinus surgery. One designer grabbed a marker, a film canister, and a clothespin and taped them together. “Do you mean like this?” he asked.



Design Thinking Process: An Inspiring Summary Video!



LINK: https://www.youtube.com/watch?v=_r0VX-aU_T8

Human Centered Design ► Hope You Get Inspired! 😊



LINK: <https://www.youtube.com/watch?v=musmgKEPY2o>

Design Thinking Quotes...

Design Thinking is a mindset.

Design thinking is about believing we can make a difference, and having an intentional process in order to get to new, relevant solutions that create positive impact.

Design Thinking gives you faith in your creative abilities and a process for transforming difficult challenges into opportunities for design.

It's Collaborative. Several great minds are always stronger when solving a challenge than just one. Design Thinking benefits greatly from the views of multiple perspectives, and others' creativity bolstering your own.

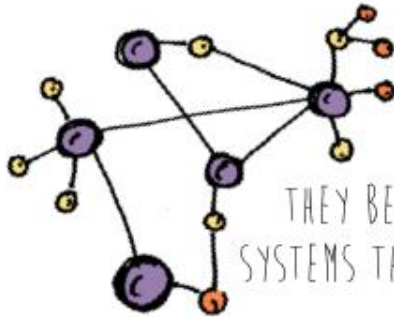
It's Experimental. Design Thinking gives you permission to fail and to learn from your mistakes, because you come up with new ideas, get feedback on them, then iterate.

It's Human-Centered. Design Thinking begins from deep empathy and understanding of needs and motivations of people

It's Optimistic. Design Thinking is the fundamental belief that we all can create change—no matter how big a problem, how little time or how small a budget. No matter what constraints exist around you, designing can be an enjoyable process.



THEY GROW MORE
EMPATHETIC



THEY BECOME
SYSTEMS THINKERS.

THEY BECOME
HACKERS AND
REBELS



THEY BECOME EXPLORERS.



THEY BECOME WILDLY AND
UNABASHEDLY DIFFERENT



THEY BECOME
PROBLEM-SOLVERS

WHAT HAPPENS WHEN STUDENTS EMBRACE DESIGN THINKING?

BY JOHN SPENCER @SPENCERIDEAS



THEY LEARN TO THINK DIVERGENTLY
(THINKING OUTSIDE THE BOX BY THINKING DIFFERENTLY
ABOUT THE BOX)



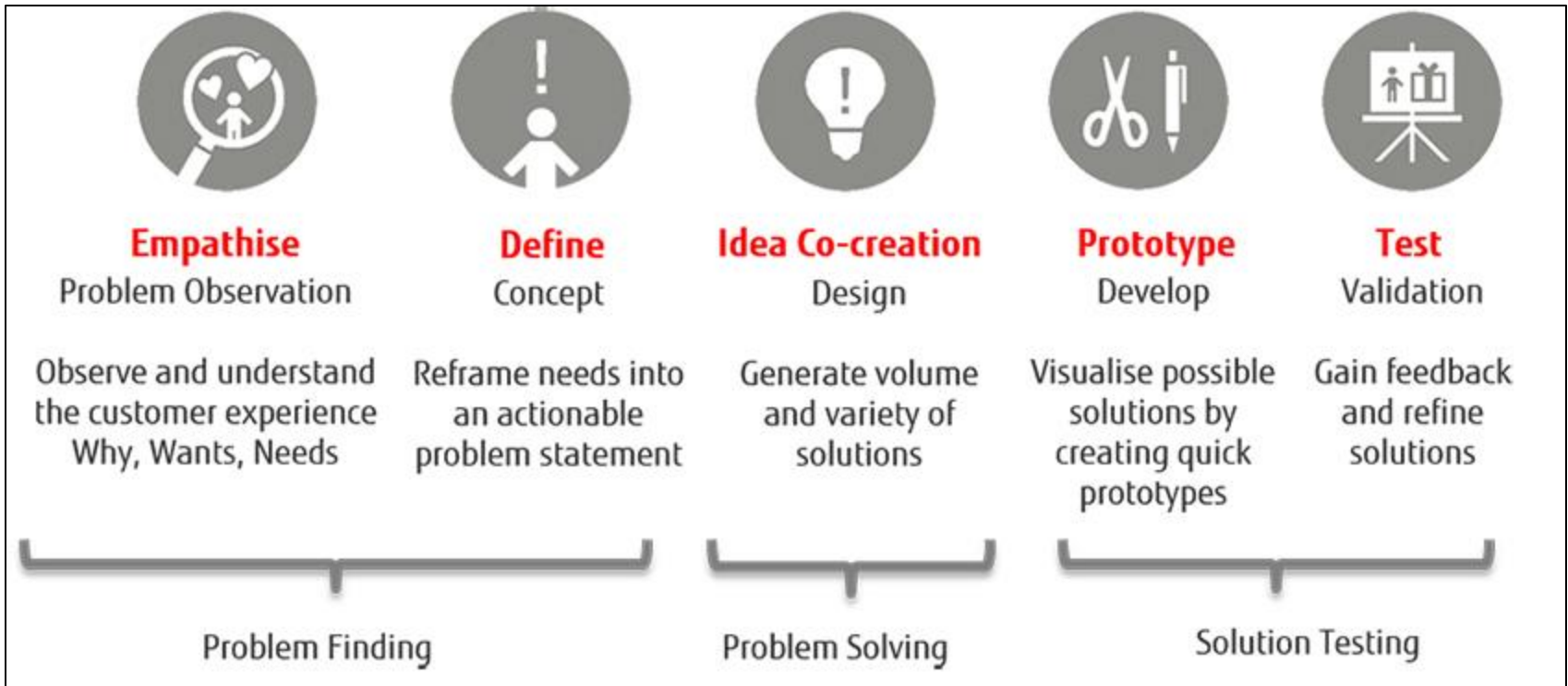
THEY MAKE CONNECTIONS
BETWEEN IDEAS.



THEY ARE READY FOR THE
CREATIVE ECONOMY



THEY LEARN TO TAKE
CREATIVE RISKS.



Thank You!
Term 4, EPD
March 2026